

Roll No.

--	--	--	--	--	--	--

ODD END SEMESTER EXAMINATION DECEMBER - 2025

Name of the Course: B.Tech. (Aerospace Eng.)

Semester: V

Name of the Paper: Control Engineering

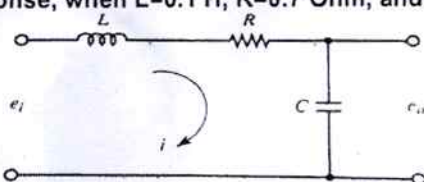
Paper Code: **TAS 511**

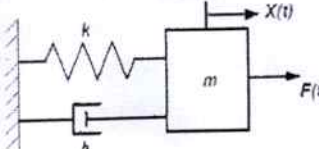
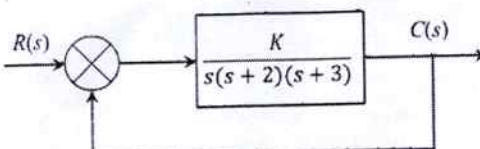
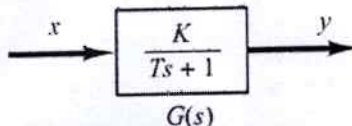
Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	Describe the open-loop and closed-loop systems with neat block diagrams? Also, discuss the advantages of a closed-loop system.	CO1
(b)	Find the solution $x(t)$ of the following differential equation: $\ddot{x} + 4\dot{x} + 5x = 7 ; \text{ where } x(0) = 1 \text{ and } \dot{x}(0) = 0$	
(c)	Plot and write the expressions (with Laplace transform) for the following signals: step, ramp, sinusoidal, and impulse.	
Q2	(20 marks)	
(a)	What is the transfer function of a system? Also, discuss about the stability in terms of poles of a system.	CO2
(b)	Linearize the following nonlinear equation around an operating point of $x=2$. $y = 5x^2 + 3x - 7$ Also, check the error, if the operating point is changed to $x=3$.	
(c)	Derive the transfer function of the following electrical R-L-C circuit. Consider input is e_i and output is e_o . Also, discuss about the stability of the system's response, when $L=0.1$ H, $R=0.7$ Ohm, and $C=0.01$ F. 	
Q3	(20 marks)	
(a)	Derive the expression for the unit-step response of a critically damped second-order system.	CO3
(b)	Comment about the stability of a system using Routh's stability criterion, whose characteristic equation is given by the following polynomial: $s^4 + s^3 + 4s^2 + 2s + 7 = 0$	
(c)	Describe the following transient-response characteristics of a control system: (i) Delay time (ii) Rise time (iii) Peak time (iv) Maximum overshoot and (v) Settling time.	

Q4	(20 marks)		CO4
(a)	What do you understand by the modern control theory? Write the standard state space equations for a linear and nonlinear dynamic system.		
(b)	Derive the state space model of the mass-spring-dashpot system shown in the figure below. 		
(c)	Plot the root locus of the following closed-loop system. 		
Q5	(20 marks)		CO5 & CO6
(a)	Describe the Proportional-Integral-Derivative (PID) control action with governing equations.		
(b)	Obtain the steady state output y of the system shown in the figure below for a given sinusoidal input $x = A\sin(\omega t)$. 		
(c)	Draw Bode plots for the integral term $G(s)=K/s$.		